

SUMMARY OF THE INVENTION

The present disclosure describes systems and methods for acquiring intraluminal distance measurements and tissue compliance characteristics using an insertable sensing device configured to operate within a body lumen under controlled measurement conditions.

In one aspect, the invention provides an intraluminal profiling device comprising an insertable measurement section, a distal sensing region configured to determine distance to an internal anatomical boundary, and a compliance characterization region configured to measure tissue response to controlled interaction.

The insertable measurement section may comprise a compliant or semi-compliant structural backbone enclosed within a body-safe outer material configured for safe insertion, positional stability, and repeatable measurement alignment. The insertable section may vary in length, geometry, flexibility, and cross-sectional profile depending on embodiment, intended anatomy, manufacturing constraints, or performance requirements.

A distal tip region of the device may incorporate ranging modalities configured to determine distance between the device and an anatomical terminus using optical time-of-flight sensing, LiDAR-based ranging, ultrasonic time-of-flight sensing, acoustic imaging methods, or combinations thereof. Certain embodiments may utilize simplified single-sensor architectures, while advanced embodiments may employ multi-point sensing or scanning approaches configured to reconstruct intraluminal geometry.

The device further comprises a compliance measurement region configured to characterize luminal deformation, pressure response, expansion tolerance, or tissue interaction behavior. Compliance characterization may be achieved using pressure sensors, capacitive sensing structures, piezoresistive elements, strain measurement systems, inflatable expansion structures, monitored pump output parameters, or hybrid combinations thereof.

In embodiments incorporating an inflatable region, controlled expansion may be produced using manual pumping mechanisms, automated pumping systems, or electronically regulated inflation architectures. Measurement of expansion behavior may be derived from pressure monitoring, strain detection, displacement measurement, flow monitoring, actuator feedback, or computational inference derived from sensor fusion.

A positional reference element may be incorporated between the insertable region and an external handle portion to establish a repeatable insertion depth reference including flanges, stops, compliant barriers, ergonomic transitions, or geometry features configured to improve measurement repeatability and user safety.

The device may further include a handle region configured to house electronic control components including processing circuitry, communication modules, power sources, pumping mechanisms, motion sensing elements, or user feedback systems.

In certain embodiments, the device operates as an independent measurement instrument capable of generating intraluminal distance and compliance metrics without reliance on external hardware, or as a coordinated sensing node within a multi-device physiologic measurement platform.

The invention further contemplates implementations incorporating verification or spoof-resistance protocols configured to confirm valid biological measurement conditions through software validation procedures, sensor confirmation mechanisms, physiologic plausibility analysis, or combinations thereof.

Data generated by the device may be stored locally, transmitted wirelessly, encrypted during transmission or storage, and processed to produce abstracted representations of physiologic characteristics suitable for privacy-preserving visualization and analysis.

The disclosed systems therefore enable repeatable acquisition of intraluminal spatial and compliance information using compact devices scalable across simplified consumer implementations, advanced imaging configurations, research applications, and integrated multi-node physiologic ecosystems.